

# Strategic Pricing and Adoption Dynamics for Hydrogen Mobility Technologies.

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## Abstract

The adoption of new technologies in heavy-duty transportation, such as hydrogen fuel cell electric vehicles (FCEVs), is shaped by upfront costs, uncertain future benefits, and strong network effects. This paper develops a dynamic model of technology adoption under uncertainty and network effects, focusing on how a hydrogen supplier decides when to enter markets and how to price or design contracts for heterogeneous trucking customers (fleet operators). We calibrate the model using two datasets: New York’s Advanced Clean Fleets (ACF) reporting data and Bizminer industry profiles of New York trucking operators. These calibrations allow us to estimate demand, simulate market outcomes, and evaluate the effectiveness of subsidies targeting station construction and technology adoption costs. The results show that personalized contracts significantly increase the profits of the hydrogen manufacturer compared to single-price benchmarks. We find that hydrogen station subsidies accelerate network formation, while customer-side subsidies expand adoption once the technology is present in a market.

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# 1 Introduction.

The transition to carbon-neutral technologies in the mobility sector presents a complex strategic challenge for firms. Companies offering greener alternatives such as hydrogen fuel cell electric vehicles (FCEVs) face challenges when pricing their goods to accelerate technology adoption. Pricing a new technology is particularly challenging due to the heterogeneity of the willingness to pay of potential customers, resulting from varying environmental concerns, regulatory pressures, and costs. More generally, the adoption of new technologies is driven by a series of individual decisions, each influenced by a comparison of uncertain benefits and uncertain costs. Potential adopters must weigh not only whether to adopt but also when to adopt, balancing the option of early adoption against the potential advantage of deferring the decision until later, when the technology is more established and potentially cheaper.

A key driver of these adoption decisions is the presence of network effects, which can significantly accelerate the spread of new technologies. For example, the attractiveness of owning a fuel cell electric vehicle (FCEV) depends on the availability of charging infrastructure. The more charging stations that are available, the more appealing it becomes to own an FCEV, thereby increasing demand for the vehicles. This growing demand, in turn, incentivizes further investment in charging infrastructure, creating a feedback loop that enhances both the supply and attractiveness of the technology. Furthermore, the benefits of adopting a new technology are realized over time, as ongoing "flow" benefits throughout the life of the innovation. In contrast, the costs of adoption are often incurred upfront and are largely irreversible. So, prior to adoption, individuals face a question: Do the anticipated long-term benefits justify the fixed costs of adoption?

While the decision to adopt lies ultimately with consumers, it is influenced by the actions of both suppliers and policymakers. Suppliers may alter the costs and benefits of adoption through pricing strategies, product improvements, or the creation of complementary goods or services. Identifying actionable customer characteristics, such as fleet size, type of load and routes, is vital for hydrogen firms to tailor their offerings and maximize adoption. Policymakers, on the other hand, can intervene to shape the technology landscape through subsidies, regulations, or tax incentives. For instance, subsidies that lower the initial costs of adopting electric or hydrogen-powered vehicles or promote the construction of charging stations can shift the cost-benefit calculus in favor of earlier adoption.

In this paper, we introduce a formal model to capture the dynamics of technology adoption under conditions of uncertainty, network effects, and policy interventions. The model incorporates individual decision-making under incomplete information, the role of network complementarities, and the influence of supply-side strategies and government policies on adoption rates. Our objective is to provide insight into the factors that drive technology diffusion and the conditions under which adoption accelerates or stagnates. The model also allows for the examination of policy effectiveness in promoting new technologies, particularly in industries characterized by high upfront costs and strong network effects, such as electric and hydrogen mobility.

We use reporting data from the Advanced Clean Fleets Act on the State of New York and Bizminer industry profile data on truck operators to estimate fuel demand of truck operators and we calibrate and run simulations of our model. Our results demonstrate the significant profit increase advantages of personalized contracts over a single market-level price, showing a substantial improvement in profits for the hydrogen manufacturer and larger market shares. For instance, while the benchmark single-price scenario limits firms to serving only a fraction of the markets with minimal penetration, the personalized contract approach enables full market entry and increases profitability.

Our analytical results characterize the optimal pricing and adoption strategies in the market with network effects. The model highlights a fundamental tradeoff faced by hydrogen suppliers: whether to enter markets early to capitalize on network effects despite high initial costs, or to delay entry until technology costs decline. The optimal solution balances the loss of today’s flow revenue against potential future cost advantages. Our analysis demonstrates that when consumer distributions vary across markets, the optimal entry strategy depends critically on both market size and the shape of distribution tails: markets with more heavy-tailed distributions of high-demand customers should be developed earlier.

Without additional assumptions, the entry strategy of the hydrogen firm and the participation conditions for the truck operators cannot be computed analytically. Instead, we numerically solve for the equilibrium strategies. We use reporting data from the Advanced Clean Fleets Act on the State of New York and Bizminer industry profile data on truck operators to estimate fuel demand of truck operators and calibrate our model. Numerical simulations show that in some cases, entering markets at an initial loss proves optimal when the network effects generate sufficient cumulative profits over time. Our results also demonstrate the significant advantages of personalized contracts over a single market-level price, showing up to 20 times higher profits and substantially greater market penetration (47% versus 9.4%). The simulations reveal that station subsidies expedite market entry timing without significantly affecting final market share, while vehicle subsidies directly expand adoption by covering customer-side switching costs. We find that high-demand customers (requiring 40+ tons of hydrogen annually) initially contribute disproportionately to supplier profits, but this concentration decreases over time as network effects enable profitable service to lower-demand segments. Overall, our findings suggest that a sequential approach to hydrogen mobility adoption with coordinated supply-side incentives early in the adoption cycle accelerates adoption and outperforms single-price strategies.

## 2 Literature Review

Our paper contributes to the existing literature on hydrogen mobility markets and strategic contracting under network effects. We organize our literature review into four main strands that inform our analysis of the hydrogen market. The first strand examines the patterns of technology adoption in transportation markets. The second strand analyzes the development of the hydrogen market for mobility, including challenges

in infrastructure deployment and competition with electric vehicles. The third strand covers strategic pricing and contracting mechanisms in emerging technology markets, focusing on optimal contract design under uncertainty. The final strand evaluates policy effectiveness and regulatory frameworks, particularly zero emission vehicle mandates and subsidy design optimization.<sup>1</sup>

**Technology Adoption and Policies in Transportation Markets.** The transportation technology adoption provides information on the mechanisms driving the adoption of alternative fuel vehicles. [Katz and Shapiro \(1986\)](#) seminal work on network externalities establishes that technologies with network effects face coordination problems requiring critical mass for successful adoption. This theoretical foundation proves particularly relevant for hydrogen vehicles, where refueling infrastructure and vehicle adoption exhibit strong mutual dependence. Most recent work has focused on the market for electric vehicles, showing that developing the network of charging stations early is the best way of accelerating adoption. [Li et al. \(2017\)](#) demonstrates that network effects between electric vehicle sales and charging stations create feedback loops that account for 40% of the impacts of federal incentives. [Muehlegger and Rapson \(2023\)](#) provide evidence that electric vehicle subsidies achieve almost 100% pass-through rates when properly targeted, their findings indicate that early adopters generally exhibit higher income, environmental awareness, and familiarity with technology.

The policy literature examining clean vehicle adoption reveals significant variation in instrument effectiveness and design features. Zero emission vehicle (ZEV) mandates consistently demonstrate superior cost-effectiveness compared to other policy approaches ([Melaina et al., 2021](#)) and that penalty structures significantly affect outcomes ([Helveston et al., 2021](#)). The effectiveness of infrastructure versus vehicle subsidies shows clear patterns that favor supply-side interventions. [Li et al. \(2017\)](#) find that infrastructure subsidies generate 2.3 times higher adoption multipliers than vehicle purchase incentives (\$2,100 vs \$6,800 per additional vehicle).

California’s Advanced Clean Trucks and Advanced Clean Fleets regulations emphasize infrastructure development alongside vehicle mandates. The Advanced Clean Trucks regulation literature reveals important spillover effects and market dynamics. Analysis by [California Air Resources Board \(2020\)](#) shows that mandate states capture disproportionate shares of zero emission vehicle supply despite smaller overall market shares, creating availability constraints in non-mandate states. European hydrogen policy framework analysis provides comparative insights into subsidy design optimization. [Wappler et al. \(2022\)](#) demonstrate that production subsidies prove more effective than demand-side incentives for early technology development, though industrial applications can crowd out transportation demand without careful sector coordination.

**Hydrogen Market Development.** The hydrogen mobility literature reveals significant regional variations in the strategies and outcomes of market development. Globally, South Korea and the United States lead in fuel cell vehicle adoption, with passenger cars dominating 82% of current deployments ([Robinius et al., 2018](#)). However,

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<sup>1</sup>For a more comprehensive study on hydrogen mobility see [Basulto et al. \(2024\)](#).

heavy-duty applications demonstrate the strongest commercial potential due to range requirements and predictable usage patterns.

Some recent work has focused on understanding the demand for hydrogen mobility considering network effects. [Grüger et al. \(2019\)](#) provide a comprehensive framework for geospatial modeling of hydrogen demand, incorporating petrol station locations, public transportation operators, and fleet characteristics. Their methodology parallels our approach to analyzing California’s trucking sector, though they focus primarily on passenger vehicle applications. The spatial dimension proves critical for understanding infrastructure deployment and network effects in hydrogen markets. However, infrastructure development faces the classic chicken-and-egg problem between vehicle adoption and refueling station availability. To address this challenge, [Kurtz et al. \(2020\)](#) demonstrate that stochastic demand modeling can improve infrastructure planning, while [Liu et al. \(2012\)](#) show that coordinated development strategies reduce overall system costs. Network optimization studies consistently find that minimum critical mass requirements are necessary for meaningful network effects to emerge. In this paper, we build on these studies by endogenizing the network effects on hydrogen demand and analyzing how hydrogen firms capture this demand through long-term contracts when deciding optimal station deployment strategies.

Market projections exhibit substantial uncertainty, with demand estimates ranging from 18.8 million to 381.3 million tonnes by 2050 across different scenarios ([Elgowainy et al., 2020](#)). This uncertainty reflects both technology development risks and policy uncertainty, highlighting the importance of adaptive strategies and robust planning frameworks such as the ones presented in this paper. Competition with electric vehicles also limits hydrogen technology adoption. [Yusaf et al. \(2022\)](#) show fuel cell vehicles scoring higher on multi-criteria performance metrics, particularly for range and energy efficiency, again highlighting the possibility for the adoption of hydrogen vehicles for trucking operations. However, TCO costs remain 1.2-12.1 times higher than conventional vehicles, creating significant adoption barriers.

**Strategic Pricing and Contracting in Technology Markets.** The strategic pricing literature provides frameworks for understanding optimal contracting and pricing mechanisms in emerging technology markets. In early work, [Aghion and Howitt \(1994\)](#) establish that optimal contracting under uncertainty requires front-loaded incentives that decrease over time as adoption occurs. This theoretical insight directly applies to hydrogen fuel supply contracts where demand uncertainty and network development create complex contracting challenges. [Kleindorfer and Wu \(2003\)](#) analyze long-term contracting with spot market integration, finding that hybrid mechanisms combining guaranteed capacity access with flexible pricing create efficient allocation. Their framework applies directly to hydrogen fuel supply, where infrastructure investments require long-term commitments while operational flexibility remains valuable for fleet operators. However, these studies focus primarily on general contracting principles without explicitly modeling the heterogeneous customer base that characterizes hydrogen mobility markets. In contrast to this literature, our approach explicitly incorporates customer heterogeneity and regulatory interventions, and we try to answer how hydrogen suppliers can leverage observable customer characteristics to design dif-

ferentiated pricing strategies.

The uncertainty literature reveals that demand volatility affects optimal pricing strategies differently depending on cost structures and market characteristics (Anderson and de Palma, 2001, Murphy and Smeers, 2015). Government intervention fundamentally changes these dynamics, making risk-sharing contracts more valuable under uncertainty. While existing work assumes homogeneous demand responses, we analyze how fleet size, operational patterns, and regulatory constraints create distinct customer segments that warrant tailored contracting approaches in the hydrogen mobility context.

### 3 Model.

We examine an economy with  $\mathbf{M} = \{1, \dots, M\}$  different markets or locations, each one has a mass  $\phi_m$  of heterogeneous consumers, indexed by a vector of characteristics  $\theta$  with continuous distribution  $f_m(\theta)$ . Time is discrete and finite  $t = 1, \dots, T$ . At each time  $t$ , a firm that produces an alternative fuel - hydrogen - makes two decisions: 1) whether to build a new station at a cost  $C_{mt}$  for location  $m$  and 2) which contracts  $\{(q_{mr}(\theta), p_{mr}(\theta))\}_{r \geq t}$  to offer different consumers. The cost  $C_{mt}$  captures the investment of constructing this station, which could partially be covered by a government subsidy.

The contracts specify fuel quantities  $q_{mr}(\theta)$  that the company promises to supply to customers of type  $\theta$  in market  $m$  at a price  $p_{mr}(\theta)$ , from the moment they contract onward. Let  $N_t$  the number of markets that the firm serves at time  $t$ . The consumers then decide whether to switch to the alternative fuel at an initial cost  $P_t$  and contract with the supplier, or continue using their current technology and receive certain utility  $\bar{u}_{mt}$ . Consumer utility (profit) from switching to the alternative fuel at time  $t$  and contracting with the supplier is given by

$$u_{mt}(\theta) = V_{mt}(q_{mt}(\theta), N_t, \theta) - P_t - \sum_{r=t}^T q_{mr}(\theta) p_{mr}(\theta) + \sum_{r=t+1}^T E_{N_r}[V_{mr}(q_{mr}(\theta), N_r, \theta)]$$

Where  $V_{mt}$  is the flow benefit of owning the alternative fuel vehicle when the network size is  $N$ .

Let  $1_{mt} = 1$  if the supplier firm enters market  $m$  at time  $t$  and 0 otherwise. And  $A_{mt} = 1$  if the firm is active in market  $m$  at time  $t$ . We can write down the firm's

problem at time  $t = 1$  as:

$$\max_{\{1_{mt}, (q_{mt}(\theta), p_{mt}(\theta))\}_{t=1, \dots, T}} \left\{ \underbrace{\sum_{t=1}^T \sum_{m=1}^M \int A_{mt} q_{mt}(\theta) (p_{mt}(\theta) - c_t) f_m(\theta) d\theta}_{\text{Flow profits}} - \underbrace{\sum_{t=1}^T \sum_{m=1}^M 1_{mt} C_{mt}}_{\text{Initial fixed cost}} \right\} \quad (1)$$

$$\text{s.t. } u_{mt}(\theta) \geq \bar{u}_{mt}(\theta), \quad \forall \theta, m, t \quad (2)$$

$$u_{mt}(\theta) \geq E[u_{m\tau}(\theta)], \quad \forall \theta, m, \tau \geq t \quad (3)$$

$$N_t = \sum_{m=1}^M A_{mt}; \quad A_{mt} = \sum_{\tau=1}^t 1_{m\tau} \quad (4)$$

The firm's flow revenue is given by the first term in equation (1), it comes from aggregating the profit from each contract in all markets where the firm is active. The fixed cost term considers the costs of building an station at the time the company decides to enter the market. Equation (2) is the buyer's participation constraint, the buyer switches to hydrogen technology at time  $t$  only if it expects to have higher total profits from switching than those of an outside option  $\bar{u}$ . Condition (3) requires that consumers switch at period  $t$  instead of waiting for one more period only if they expect to have higher profits from adopting. Finally, equation (4) is a consistency condition, the size of the network at time  $t$  is the sum of all markets where the firm is active, the firm is active in market  $m$  at time  $t$  ( $A_{mt} = 1$ ) if it entered the market at time  $t$  or before.

We can write this problem recursively as

$$\Pi_t^*(A_{mt-1}, \{(q_{mr}(\theta), p_{mr}(\theta))\}_{r=t-1, \dots, T}) = \max_{1_{mt}, \{(q_{mr}(\theta), p_{mr}(\theta))\}_{r=t, \dots, T}} \left\{ \underbrace{\sum_{m=1}^M \int A_{mt} q_{mt}(\theta) (p_{mt}(\theta) - c_t) f_m(\theta) d\theta}_{\text{Flow revenue}} - \underbrace{\sum_{m=1}^M 1_{mt} C_{mt}}_{\text{Initial fixed cost}} \right\} + \Pi_{t+1}^*(A_{mt}, \{(q_{mr}(\theta), p_{mr}(\theta))\}_{r=t, \dots, T})$$

subject to

$$u_{mt}(\theta) \geq \bar{u}_{mt}(\theta), \quad \forall \theta, m$$

$$u_{mt}(\theta) \geq E[u_{mt+1}(\theta)],$$

$$N_t = \sum_{m=1}^M A_{mt}$$

In the former recursive formulation, the relevant state variables are the set of markets in which the firm was active at time  $t - 1$  and the contracts it has to supply to existing customers. It decides the new markets to enter and new contracts to sign.

We now make two assumptions on the costs. First, the cost of entering a market  $C_{mt}$  is weakly decreasing over time. Second, the initial cost of the new technology  $P_t$  for

mobility customers is weakly decreasing in expectation time, that is,  $E(P_{t+1}|P_1, \dots, P_t) \leq P_t$ . In this way, we capture the incentives that the supplier is willing to delay entry in some markets because it will face lower costs. However, this will imply a loss of revenue for some periods. Customers also have an incentive to adopt the technology later rather than sooner, as they could face lower costs and uncertainty, especially for consumers with a lower valuation of hydrogen technology. Long-term contracts will serve as a mechanism to decrease the uncertainty faced by the adopters of the technology.

## 4 Analytical Solution.

In this section, we provide partial analytical solutions to the firm's entry problem under different parametric assumptions. We assume that the firm can contract with each consumer separately and offer them quantities and prices in a long-term agreement. We take this case as a benchmark to our study and compare it to three other sets of assumptions. First, a case where the firm can only charge a single price to all participants on the market. Secondly, a case where the firm can not observe consumer characteristics, and finally a case where policy interventions (described later) are uncertain for all participants in the market.

### 4.1 Personalized Long-Term Contracts.

Now, we turn to the case where the supplier firm can offer personalized long-term contracts to each consumer in the market. The contracts  $\{(q_{mr}(\theta), p_{mr}(\theta))\}_{r \geq t}$  specify quantities and unit prices for hydrogen for each period and can depend on the relevant characteristics or 'type' of the consumer.

**Proposition 1.** Suppose consumers have rational expectation in equilibrium, i.e.

$$E_{N_r}[V_{mr}(q_{mr}^*, N_r^*)] = V_{mr}(q_{mr}^*, N_r^*)$$

And suppose the outside option  $\bar{u}_{mt} = 0$ . If the firm enter a market  $m$  at time  $t$ , it will offer contract to consumer  $i$  whenever

$$V_{mr}(q_{mt}^*, N_t^*, \theta) - P_t + E[P_{t+1}] \geq 0 \quad (5)$$

The optimal contract  $\{(q_{mr}^*, N_r^*)\}_{r \geq t}$  satisfies

$$\frac{\partial}{\partial q} V_{mr}(q_{mr}^*(\theta), N_r^*, \theta) - c_r = 0$$

And

$$p_{mt}^* = \frac{V_{mt}(q_{mt}^*, N_t^*, \theta) - P_t}{q_{mt}^*(\theta)},$$

$$p_{mr}^* = \frac{V_{mr}(q_{mr}^*, N_r^*, \theta)}{q_{mr}^*(\theta)}, \quad r > t$$

The buyer accepts the contract.

The result can be interpreted intuitively: optimal contracts equate the flow marginal revenue with marginal costs. When deciding whether to offer a contract at time  $t$  or defer it by one more period, the firm forgoes today's flow revenue,  $V_{mt}(q_t^*, N_t^*)$ , but can extract additional revenue tomorrow,  $P_t - E[P_{t+1}] > 0$ . As long as the loss of today's flow revenue exceeds the potential gains from lower technology costs in the future, the firm prefers to offer contracts early.

Contracts are able to price discriminate inter-temporally but also across consumers. Then, even when the firm makes the initial investment in one market, it might not want to offer all consumers in that market a contract and would rather defer the adoption until the switch to the new technology is cheaper for some customers or the network effects make the switching more attractive for consumers with lower valuation for the fuel.

**Proposition 2.** If  $V_{mt}(\cdot)$  is increasing in  $q$  and  $\theta$ , then there exists some  $\underline{\theta}_t$  such that  $V_{mt}(q, N, \underline{\theta}_t) - P_t + P_{t+1} = 0$  and  $\forall \theta \geq \underline{\theta}_t$   $q^* > 0$ , i.e. the seller offers a contract to all customers above a certain threshold.

With propositions 1 and 2, the profit from entering market  $m$  at time  $t$  is given by

$$\Pi_t^m := \sum_{\tau=t}^T \pi_\tau^m$$

With

$$\pi_t^m := \phi_m \int_{\underline{\theta}_t}^{\infty} (V_{mt}(q_{mt}^*(\theta), N_t^*) - q_{mt}^*(\theta)c_t) f_m(\theta) d\theta - P_t \phi_m (1 - F_m(\underline{\theta}_t)) \quad (\text{rt1})$$

$$\pi_\tau^m := \phi_m \int_{\underline{\theta}_\tau}^{\infty} (V_{m\tau}(q_{m\tau}^*(\theta), N_\tau^*) - q_{m\tau}^*(\theta)c_t) f_m(\theta) d\theta - P_{t+1} \phi_m (F_m(\underline{\theta}_{\tau-1}) - F_m(\underline{\theta}_\tau)), \quad \tau > t \quad (\text{rt2})$$

Solving the firm's entry problem is complex, using the recursive formulation of the problem, if at any time  $t$  the firm can choose  $K$  markets to enter then we will have to check  $2^K$  possible combinations of markets (e.g. enter all markets, enter only market 1, enter market 1 and 3, enter market 3,4,5, etc.). Then, we have to carry back all the  $2^K$  optimization problems backwards in time in order to give the continuation value at time  $t - 1$ . Despite this computational challenge, we can establish some principles about optimal entry strategy. When all markets have identical size ( $\phi_1 = \phi_2 = \dots = \phi_M$ ), if one market's consumer distribution first-order stochastically dominates (FOSD) the others, the firm optimally enters that market first. In contrast, when consumer distributions are identical across markets, the firm prioritizes entering the largest market. In the general case, where both market sizes and distributions vary, optimal entry depends on a combination of market size and the shape of the distribution tails.

## 4.2 Computation.

The use of computational methods is essential for solving finite-time dynamic contracting models, particularly when the state space is large and the problem lacks closed-form

solutions. Analytic approaches would require deriving the optimal actions at every state in this high-dimensional space, which quickly becomes intractable due to the complexity introduced by the dynamics between network effects and the change of future costs and incentives. To solve our model computationally, we would follow the following.

**Algorithm 1.**

1. At time  $T$ , compute all possible 'active' market combinations  $\sigma(M, T)$  (e.g with 2 markets  $(0, 0), (0, 1), (1, 0), (1, 1)$ ).
2. For each 'state'  $\sigma, T$  compute  $\pi_T^m = \pi_T^m - C_{mT} * 1_{mT}$  for all markets using (rt1) and (rt2), choose the optimal entry market that maximizes the profits, save the decision and the profits ('value functions').
3. Go back to  $T - 1$  and compute  $\sigma(M, T - 1)$  and repeat step 2 (now, the value of the profits depends on where you will end up at the start of period  $T$ ).
4. Repeat for  $t = T - 2, \dots, 0$ .

In practice, we would run into trouble computing the permutations of the large state space and the integration on the profit in expressions (rt1) and (rt2). Adding uncertainty means one extra integration at each state and time. To handle the permutation we could reduce the state space by only allowing the firm to enter a certain number of markets at one time, this assumption is not problematic if the markets we are interested in have high entry costs and the supplier firm has some capacity constraints. Using certain assumptions of functional form on the parameters helps with the computation of (rt1) and (rt2).

#### 4.2.1 Utility Representation.

To model truck operators' fuel choices in the econometric approach, we employ a Conditional Choice Probability (CCP) approach following Hotz and Miller (1993). This framework is well-suited to our setting for two reasons. First, it allows us to recover value functions directly from observed choice probabilities without explicitly solving the full dynamic programming problem at each iteration, significantly reducing computational burden. Second, it exploits the natural persistence in truck operators' fuel decisions, operators rarely switch powertrains every period due to substantial switching costs. The CCP approach enables us to tractably incorporate the hydrogen manufacturer's contract design problem by computing the probability that operators accept contracts and remain with hydrogen technology conditional on their observable characteristics and the evolving network size.

Under this approach, suppose that at each time  $t = 1, \dots, T$ , each truck owner  $j = 1, \dots, J_m$  is characterized by the powertrain (fuel) that they are currently using in their truck  $F_{jt} = (D, H)$ , and a vector of static observed characteristics  $(\theta_j, X_j) = (\text{Mileage}, \{\text{ZIP}, \text{NAICS}, \text{Sust. Obj.}, \text{Network Size}\})$ . Each period, they choose which fuel to (continue to or switch to) use,  $f \in \{D, H\}$  and how much of the fuel they use

$q$  based on their characteristics and a state of the world  $S_t = (TCO_t, Policies_t)$ . Their period payoff is given by

$$u_{jt}(f, \theta_j, F_{jt}, X_j, S_t) = \underbrace{\beta_1 \theta_j + \beta_2 X_j}_{R_j(\theta_j)} - q(\theta_j)p(\theta_j) - TCO_{ft} - C_{ft} * 1\{F_{jt} \neq f\} + \epsilon_{jft} \quad (6)$$

Where  $\beta_1, \beta_2$  are revenue parameters to estimate,  $C_{ft}$  is a switching cost if the truck owner decides to switch power trains in period  $t$  and  $\epsilon_{jft}$  is a T1EV distributed by random utility shock that is independent across time, consumers and fuel type.

Note that this formulation assumes that the truck operators can always opt out of the hydrogen contract and return to a diesel truck. However, they would have to pay a switching cost. This assumption limits the hydrogen firm's pricing power as truck operators always have an outside option. The fact that the outside option comes with a switching cost implies that truck owners will not optimally switch power trains every period, matching the observed ownership patterns in the data.

Let  $V_{j,t+1}(\theta_j, F_{jt}, X_j, S_t)$  be the optimal value function, that is, the total utility that the truck owner  $j$  gets if he follows his optimal policy from period  $t + 1$  onward after choosing fuel  $F_{jt}$  in the current period. We define the conditional value function of period  $t$  as

$$v_{jf}(\theta_j, F_{jt}, X_j, S_t) = u_{jt}(f, \theta_j, F_{jt}, X_j, S_t) + V_{j,t+1}(\theta_j, f, X_j, S_t)$$

The probability of observing action  $j = H$  at time  $t$  is given by

$$\mathbf{P}_H(\theta_j, F_{jt}, X_j, S_t) = \frac{1}{1 + \exp[v_{jD}(\theta_j, F_{jt}, X_j, S_t) - v_{jH}(\theta_j, F_{jt}, X_j, S_t)]} \quad (7)$$

From [Hotz and Miller \(1993\)](#), for all  $F_{jt}, X_j, \theta_j, S_t$  the value function can be written as

$$V_{j,t+1}(\theta_j, F_{jt}, X_j, S_t) = v_{jH}(\theta_j, F_{jt}, X_j, S_t) - \ln(\mathbf{P}_H(\theta_j, F_{jt}, X_j, S_t)) + 0.57... \quad (8)$$

Suppose that at time  $\tau$  a truck operator owns a diesel truck  $F_t = D$  and is approached by the hydrogen manufacturer. The hydrogen manufacturer offers a contract that maximizes

$$q_t(\theta_j)(p_t(\theta_j) - c_t)\mathbf{P}_H(D, \theta_j, X_j, S_t) + \sum_{t=\tau}^T q_t(\theta_j)(p_t(\theta_j) - c_t)\mathbf{P}_H(H, \theta_j, X_j, S_t) \quad (9)$$

Subject to equations 7 and 8.

In equation 9, at period  $\tau$  the relevant probability is the probability that the truck owner switches to hydrogen given that he is using diesel one period before. For the later periods, the profits are realized only if the truck manufacturer prefers to keep the hydrogen truck rather than change back to diesel.

The implementation of this formulation will consist of four steps.

1. Regress the revenue on mileage and observed characteristics to retrieve  $\beta_1$  and  $\beta_2$ .
2. Get estimates of the TCO and the switching costs for hydrogen trucks to compute equation (7).
3. Compute the CCPs and the conditional value functions for all the observed trucks in the sample.
4. Solve the maximization problem of the hydrogen firm using **Algorithm 1**.

We develop a program that solves the model. This means that based on entry costs  $C_{mt}$ , marginal production costs  $c_t$ , switching costs  $P_t$ , mobility consumer-type distribution, and government incentives, we can compute optimal entry decisions, market shares, and optimal contracts. In the code, we allow the firm to enter anywhere from 1 to all  $M$  markets at one time.<sup>2</sup>

We also run simulations of two relevant policy simulations. **Policy 1: Station Subsidy.** By subsidizing the entry of the supplier firm, there are no barriers to entry so the firm enters all markets at time 1. **Policy 2: Truck cost subsidy.** Subsidizing the technology adoption costs for mobility customers does not change the entry decisions of the firm compared to the optimal case with no subsidies, however prices and market shares will be impacted by this policy.

## 5 Data and Calibration.

In this subsection, we calibrate our model using data on truck operators and projections of hydrogen mobility industry development. We incorporate well-established literature estimates and proprietary data on hydrogen production costs, mapping entry costs to expenses for new hydrogen plants and distribution centers. We relate switching costs to the difference in total cost of ownership between fuel cell electric trucks and diesel alternatives, incorporating real government incentives outlined in Basulto et al. (2024). To determine the distribution of customer types in each market, we employ maximum-likelihood estimation using two datasets: BizMiner (2024) and State of New York (2024), which provide annual fuel demand and revenue data for medium- and heavy-duty truck operators in New York. Below, we elaborate on these data sources and detail how we map empirical observations to parameters in our model.

We analyze data from the New York State Department of Environmental Conservation’s Advanced Clean Fleets (ACF) program (State of New York, 2024).<sup>3</sup> This dataset contains detailed information on vehicle fleets and facilities across New York State. It includes organizational details (legal names, types, and parent entities), vehicle characteristics (fuel type, weight class, mileage, usage patterns), facility attributes (location, ownership, fueling/charging infrastructure), and tracks whether the firm has a public

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<sup>2</sup>Code available at: [https://github.com/genarobasulto/Contracts\\_Technology\\_Adoption](https://github.com/genarobasulto/Contracts_Technology_Adoption)

<sup>3</sup>See 7 in the Appendix for a sample observation of the data set.

announcement of sustainability goals. As shown in Table 1, the dataset encompasses 5,286 trucks across 1,214 facilities, representing 190 parent organizations operating in 701 zip codes and 107 NAICS industry classifications for the years 2022 to 2024. The fleet composition includes 1,793 Class 2b-3 trucks, 1,224 Class 4-6 trucks, and 2,252 Class 7-8 trucks, with Class 7-8 heavy-duty vehicles recording substantially higher annual mileage (33,851.09 miles) compared to medium-duty trucks (approximately 14,000 miles). Average revenue varies by vehicle class, with Class 4-6 trucks generating the highest revenue per truck (\$21.46 million), followed by Class 2b-3 (\$20.52 million) and Class 7-8 (\$19.22 million).

Table 1: ATF Reporting Summary Statistics by Truck Weight Class.

| Weight Class | Trucks | Facilities | Firms | ZIP Codes | NAICS | Mileage*               | Facility Revenue <sup>+</sup> | Revenue <sup>+</sup> per truck |
|--------------|--------|------------|-------|-----------|-------|------------------------|-------------------------------|--------------------------------|
| Class 2b-3   | 1793   | 749        | 131   | 463       | 78    | 14855.07<br>(11880.37) | 853.15<br>(505.17)            | 20.54<br>(60.57)               |
| Class 4-6    | 1224   | 542        | 132   | 344       | 80    | 14251.57<br>(10842.18) | 773.68<br>(519.77)            | 21.46<br>(67.57)               |
| Class 7-8    | 2252   | 663        | 138   | 459       | 74    | 33851.09<br>(41239.09) | 598.35<br>(525.88)            | 19.22<br>(42.78)               |
| All          | 5286   | 1214       | 190   | 701       | 107   | 22789.75<br>(29842.46) | 724.06<br>(529.88)            | 20.16<br>(55.51)               |

Note: Values in parentheses represent standard deviations. \*Self reported average miles per Year.

<sup>+</sup>Self reported annual revenue in Millions of Dollars. Source: [State of New York \(2024\)](#).

Additionally, we construct a dataset from industry market profiles using reports by [BizMiner \(2024\)](#) to analyze a sample of Californian and New Yorker trucking companies. The dataset comprises 1,646 Californian trucking companies and 1,589 New Yorker companies, categorized under eight specific NAICS industry codes: Specialized Freight Trucking (4842), School and Employee Bus Transportation (4854), Charter Bus Industry (4855), Other Transit and Ground Passenger Transportation (4859), Couriers and Express Delivery Services (4921), Local Messengers and Local Delivery (4922), General Freight Trucking, Local (484110), and General Freight Trucking, Long-Distance (484121). Bizminer provides data for up to 250 competitors within each industry that are closest to the geographical center point of the state, regardless of size. For most classification codes examined, Bizminer’s universe contained fewer than 250 competitors. Each dataset entry contains detailed company information, including names, estimated market values, and annual sales figures. For our initial model calibration, we segment the trucking companies into 9 markets using the first digit of their ZIP code.

In our calibration, we define markets by grouping observations according to the first two digits of the facility ZIP code which range from 10 to 14 across New York State, see Figure 1. To map the annual fuel demand of truck operators to the parameter  $\theta$  in our model, which represents the optimal amount of hydrogen purchased under contracts according to Proposition 1 we use two approaches. For the ACF reporting data, we convert truck mileage and vehicle class information into hydrogen demand estimates using average hydrogen fuel economy rates for different truck classes given in [Basma et al. \(2023\)](#). For the Bizminer data, we combine the companies’ 2024 revenue figures with the ACF regulation dataset’s mileage-to-revenue ratios by NAICS code to

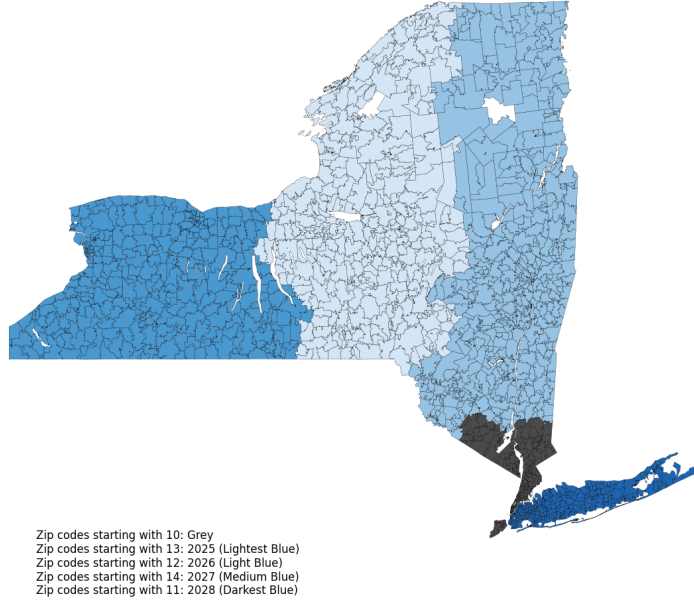


Figure 1: Relevant regions in the State of New York.

estimate mileage, which we then convert to potential hydrogen demand. Figure 2 illustrates the distribution of the annual potential demand for hydrogen across New York's five market regions using blue lines. Notice, for example, region 11 in panel (b) which mainly includes New York City, facilities there reported mostly medium-duty class 2-6 trucks (66.7%) and the range of potential hydrogen demand goes from 3 to 11 tons a year. Compare this with region 13 where 47.2% of the reported trucks are class 7-8, the demand ranges from 4 all the way to 40 tons of hydrogen a year. To calibrate the size of the network effects in the revenue of the truck operators, we run a regression of revenue on number of active active markets by firm, controlling for weight class and truck mileage. The results in Table 5 show that network effects are significant as being present in one additional market increases the average revenue of a company by around 20.5%, all else constant.

In Table 2 we regress truck revenue in millions of dollars as reported in the ACT dataset of the [State of New York \(2024\)](#) on the truck's average mileage in thousands, its weight class, the number of trucks that the truck facility operates, the fuel of the vehicle (diesel as the base category, electricity, gasoline, natural gas or propane) and region (ZIP) and industrial use (NAICS) fixed effects. The effect of mileage on revenue is positive, on average a truck brings in a revenue of around \$230-\$274 per mile. Trucks of heavier weight classes generate higher revenue and operating more trucks has a negative effect on the rentability of each one.

The region, industrial use, and mileage have a great explanatory power for truck revenue since the  $R^2$  of column (3) is 78.8%. Importantly, given the location and industrial use of the truck, the **type of fuel of the truck has an insignificant effect on revenue**. This is evidence that the revenue, which mostly comes from the use of the truck in a given location and for a given industrial activity, is fixed across observed power trains so we can assume that this would also be the case for hydrogen.

Table 2: Truck Revenue Regression.

|                         | <i>Dependent variable:</i> |                       |                       |                       |
|-------------------------|----------------------------|-----------------------|-----------------------|-----------------------|
|                         | Truck Revenue              |                       |                       |                       |
|                         | (1)                        | (2)                   | (3)                   | (4)                   |
| mileage                 | 0.230***<br>(0.009)        | 0.250***<br>(0.007)   | 0.274***<br>(0.007)   | 0.275***<br>(0.007)   |
| Class 4-6               | 1.069***<br>(0.324)        | 0.057<br>(0.288)      | 0.264<br>(0.239)      | 0.282<br>(0.216)      |
| Class 7-8               | 0.897**<br>(0.396)         | 0.030<br>(0.348)      | 0.602**<br>(0.290)    | 0.629***<br>(0.231)   |
| n_trucks                | -0.013*<br>(0.008)         | 0.008<br>(0.007)      | 0.007<br>(0.005)      | 0.007<br>(0.005)      |
| Electricity             | -1.070<br>(2.652)          | -3.676<br>(2.304)     | -3.558*<br>(1.900)    |                       |
| Gasoline                | -0.601*<br>(0.341)         | -0.050<br>(0.303)     | 0.046<br>(0.253)      |                       |
| Natural Gas             | 1.372<br>(1.725)           | 2.998**<br>(1.305)    | 2.320*<br>(1.256)     |                       |
| Propane                 | 0.852<br>(6.657)           | -3.957<br>(5.960)     | -0.196<br>(4.686)     |                       |
| Constant                | 8.301<br>(6.626)           | -12.071***<br>(1.714) | -13.033***<br>(1.363) | -13.045***<br>(1.352) |
| Observations            | 4,207                      | 4,207                 | 4,207                 | 4,207                 |
| R <sup>2</sup>          | 0.563                      | 0.612                 | 0.788                 | 0.787                 |
| Adjusted R <sup>2</sup> | 0.507                      | 0.604                 | 0.756                 | 0.756                 |
| ZIP FE                  | YES                        | NO                    | YES                   | YES                   |
| NAICS FE                | NO                         | YES                   | YES                   | YES                   |

*Note:*

\*p&lt;0.1; \*\*p&lt;0.05; \*\*\*p&lt;0.01

On the production side, the main parameters in our model are the initial costs of setting up a production and distribution center on a given location (fixed costs of production) and the variable costs of producing the hydrogen for the mobility contracts. We use proprietary data from Air Liquide along with cost estimates in the literature

(Basma and Rodríguez, 2022, Basma et al., 2023) to calibrate these parameters in the following way.

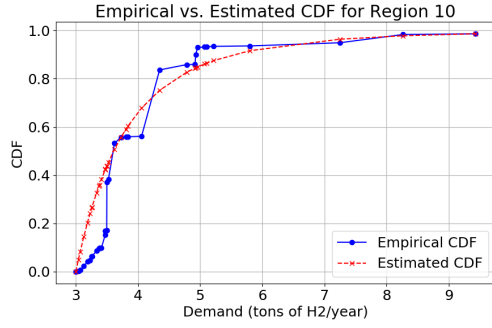
**Fixed costs.** The initial fixed costs of production are calibrated using the capital expenditure (CAPEX) for a Steam Methane Reformer (SMR) with 30 tpd capacity plus the Engineering, Procurement, and Construction costs (EPC) of building a hydrogen refueling station. The SMR assumption is based on the fact that in the US the supply chain of hydrogen for semitrucks is expected to be 100% liquid hydrogen, so we take an average capacity of 30 tpd as our baseline for calibration. The CAPEX model is based on Airliquide’s previous research (Goodhand, 2024) and adjusted for different states using construction multipliers and forecasted for future years using the Chemical Engineering Plant Cost Index<sup>4</sup> (CEPCI). Following this method we get CAPEX for a 30tpd SMR ranging from approximately \$55-65 million in Arizona and \$70-80 million in New York (2023 USD). Finally, we take the EPC costs of a typical station with capacity of 2 tons per day (approximately \$4-6 Million) and adjust the number of stations needed in each market depending on the market size.

**Variable Costs.** The variable costs of production come from the energy use of the plants, liquid hydrogen transportation costs and other operating and maintenance costs of the SMR and the refueling stations. We take the energy consumption per kg of Hydrogen of a standard SMR to be in the range of 0.6-1.3 kWh/kg H<sub>2</sub> (Oni et al., 2022). We assume that part of that renewable electricity is sourced from a Power Purchase Agreement, but renewable electricity is not always available from solar and wind farms due to the sun not shining or the wind not blowing 100% of the time. When renewable energy isn’t available from a PPA, we assume for our model that the SMRs source the rest of their electricity using a Renewable Energy Certificate. We observe the costs and what percentage of electricity comes from PPRs and RECs by US State in 2023USD/MWh so we can calculate the total variable costs of electricity for our 30tpd SMR. For SMRs, we use other costs of 4-5% of the CAPEX a year and we add operations and maintenance costs of stations accordingly (approximately 3-5% of EPC a year of maintenance and \$140,000-160,000 of operating costs in 2023, adjusted annually).

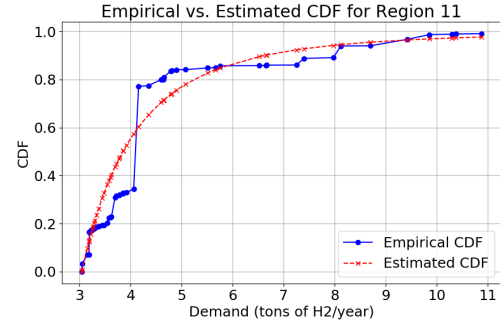
To calibrate other model parameters and justify our calibration, we draw on several key estimates from the literature. The switching costs  $P_t$  are calibrated using TCO of FCEV and diesel trucks. According to Hunter et al. (2021) and Basma and Rodríguez (2022), the total cost of ownership (TCO) differential between fuel cell electric (FCE) and diesel heavy-duty trucks, excluding fuel costs, is projected to decrease substantially—from approximately 1.14 USD/mile in 2024 to 0.16 USD/mile by 2035, primarily due to declining manufacturer suggested retail prices for FCE trucks. This projection, combined with data from our New York dataset showing that Class 7-8 trucks average between 30,879 and 43,375 annual miles and typically remain in service for 10 years, allows us to estimate that the switching cost parameter  $P_t$  will range from

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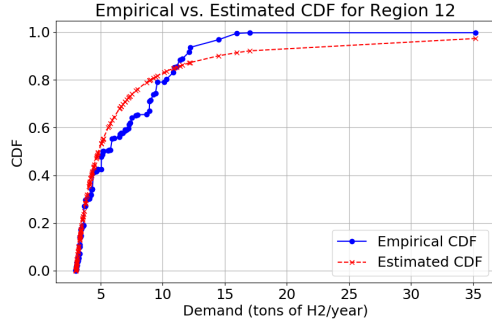
<sup>4</sup>The CEPCI consists of a composite index assembled from a set of four sub-indexes: Equipment; Construction Labor; Buildings; and Engineering & Supervision. Each index and subindex is the weighted sum of several components. Most of these components correspond to Producer Price Indexes. See <https://www.chemengonline.com/pci-home>



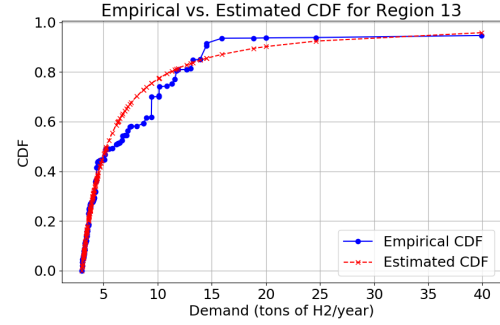
(a) ZIP starts with 10.



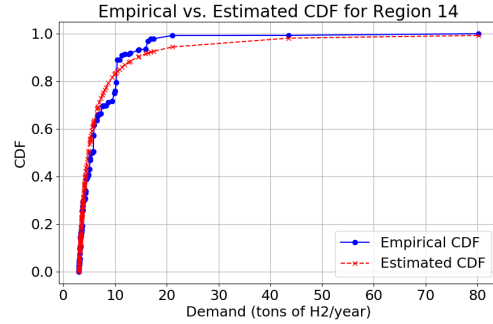
(b) ZIP starts with 11.



(c) ZIP starts with 12.



(d) ZIP starts with 13.



(e) ZIP starts with 14.

Figure 2: Distribution of Types (yearly potential demand of hydrogen) for all regions in NY state.

\$470,000 in 2025 to \$56,500 in 2035 at the highest and from \$300,000 to \$40,000 at the lowest. For hydrogen production costs, [Basma et al. \(2023\)](#) projects that technological improvements in fuel-cell truck technology will reduce green hydrogen production costs from \$7.00-\$12.00 per kg in 2022 to \$4.00-\$7.00 per kg by 2035, enabling us to calibrate our cost parameter  $c_t$ . Finally, for hydrogen infrastructure investments, estimates from [Basma and Rodríguez \(2022\)](#) and [Liu et al. \(2016\)](#) indicate that establishing a hydrogen station requires capital investments ranging from \$10 million to \$40 million, depending on station capacity.

## 6 Numerical Results.

### Based on older version of the model.

We numerically solve the model for a range of calibrations based on the parameter estimates in table 6 for a total of 72 simulations over a 10-period horizon, allowing the hydrogen firm to enter up to 3 out of 5 markets simultaneously. First, we compare the model’s optimal results using personalized contracts to a benchmark scenario with a single market-level price. Table 3 summarizes the cumulative profits, average prices across all 10 periods, the number of active markets, and market shares at the end of the 10-year horizon.

Table 3: Market outcomes under different regimes.

| Variable                      | Single Price           | Personalized Contracts | Entry Subsidy           | Truck Subsidy           |
|-------------------------------|------------------------|------------------------|-------------------------|-------------------------|
| Profits (M. USD)              | 1,023.97<br>(1,334.63) | 9,131.03<br>(6,187.30) | 18,780.23<br>(6,187.69) | 14,193.31<br>(6,796.45) |
| Prices (\$/KgH <sub>2</sub> ) | 21.36<br>(0.71)        | 9.10<br>(2.37)         | 11.12<br>(2.37)         | 18.13<br>(3.13)         |
| Active Markets                | 2-3                    | 4-5                    | 5                       | 4-5                     |
| Market Share                  | 9.41%<br>(0.013)       | 47.00%<br>(0.077)      | 47.01%<br>(0.077)       | 100%<br>(0.00)          |
| Parameter Combinations        | 24                     | 24                     | 12                      | 12                      |

Note: Values in parentheses represent standard deviations.

The results in Table 3 reveal that the profits under a single market price are only about 11% of those achieved with personalized contracts, even though the average prices are more than twice as high. In the benchmark case of a single market price, the firm optimally enters 2-3 markets, reaching an average market share of 9.41%. In contrast, with personalized contracts, the firm enters 4-5 markets and captures 47% of the market share. Figure 3 panel (a) shows the market shares over time, where personalized contracts allow the firm to leverage network effects early, enabling it to offer lower prices and achieve greater market penetration compared to the single price benchmark. The simulated profits over time are shown in Figure 3 panel (b), from the start, the firm can capture more clients using long-term contracts, so profits are higher.

We incorporate government incentives into our model, focusing on the two subsidies described in Section 4.2: a **station subsidy** and a **truck subsidy**. The results of these subsidies are presented in the last two columns of Table 3. The station subsidy does not affect the final market share, as the contracts offered are independent of the firm’s entry decisions, conditional on the size of the network i.e. the number of stations. However, it does increase profits due to the financial support provided by the subsidy. Notice however, that the profits at initial periods are still lower than those at later periods as the firm still has to incentivize the customers to switch to the technology by covering some of the adoption costs. The total profits for the firm increase by the amount of the subsidy.

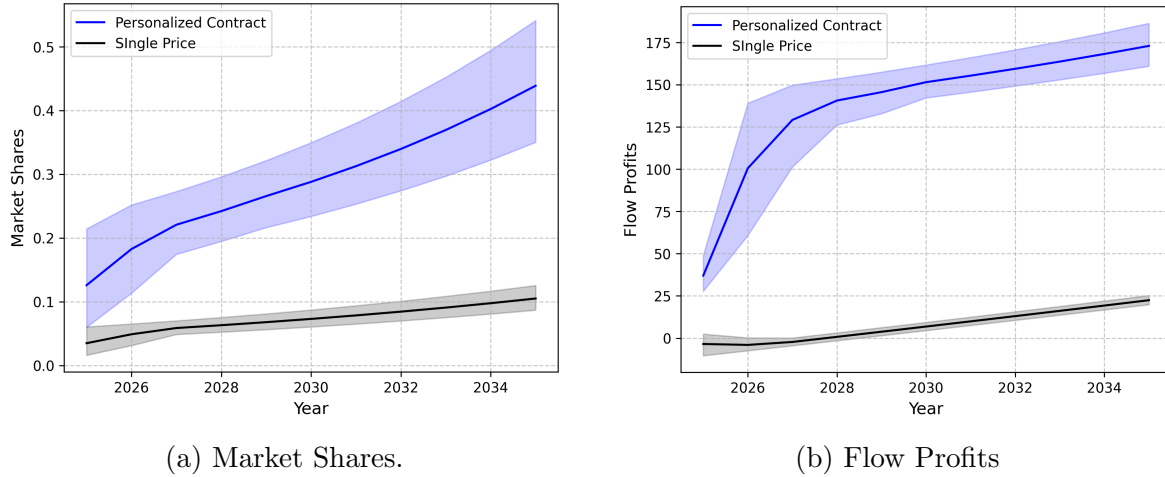


Figure 3: Projected Market Shares, Profits and Average Prices.

In contrast, the truck subsidy directly impacts market shares, as lower entry costs encourage more mobility clients to switch to hydrogen-powered vehicles, leading to higher market shares, reaching up to 100% by the end of the simulation horizon. Subsidizing the technology adoption costs for mobility customers does not change the number of markets entered by the firm compared to the optimal case with no subsidies, as the firm still enters 4-5 markets. However, profits for the firm increase with this subsidy once the firm is active in a market, since it can offer contracts to all consumers since switching costs are zero.

Table 4: Year of Entry by Market.

| Region<br>ZIP | Year of Entry |            | Initial Profit |             | Total Profit |              |
|---------------|---------------|------------|----------------|-------------|--------------|--------------|
|               | Pessimistic   | Optimistic | Pessimistic    | Optimistic  | Pessimistic  | Optimistic   |
| 10            | 2026          | 2027       | \$6.90MM       | \$18.97MM   | \$220.63MM   | \$2,336.60MM |
| 11            | 2027          | 2028       | (\$4.44MM)     | \$67.70MM   | \$189.77MM   | \$3,917.90MM |
| 12            | 2030          | 2031       | (\$5.21MM)     | \$58.76MM   | (\$677.05MM) | \$3,773.66MM |
| 13*           | -             | 2032       | -              | \$54.1448MM | (\$615.19MM) | \$3,707.44MM |
| 14            | 2028          | 2029       | \$6.91MM       | \$35.41MM   | \$337.05MM   | \$5,149.34MM |

Note: A dashed line indicates that it is optimal not to enter the market. Values based on 500 simulations.

#### Based on New Version of the model.

The results of our model provide several strategic insights for hydrogen suppliers and policy makers looking to accelerate FCEV adoption in heavy-duty transportation. Personalized contracting strategies significantly outperform uniform pricing approaches. As demonstrated in our simulations, optimal personalized contracts generate approximately 20 times the profit of single-price strategies while allowing higher market penetration (47% versus 9.4% market share). This suggests that hydrogen suppliers should prioritize developing customer segmentation and long-term relationships with customers rather than relying on market-wide pricing. Furthermore, early suppliers should expect

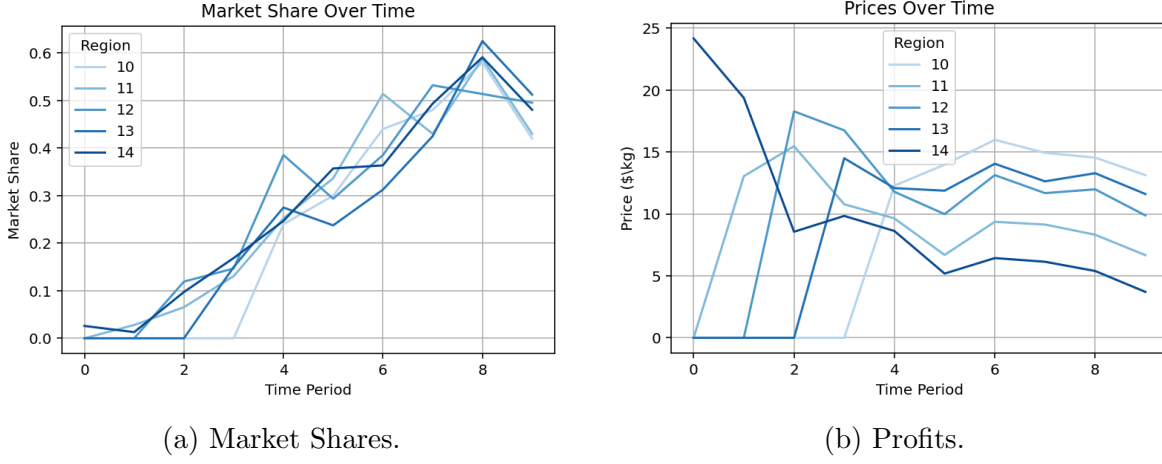


Figure 4: Projected Market Shares and Average Prices by Market.

lower and even negative margins in the beginning and only see profits over time as technology costs decline and network effects strengthen.

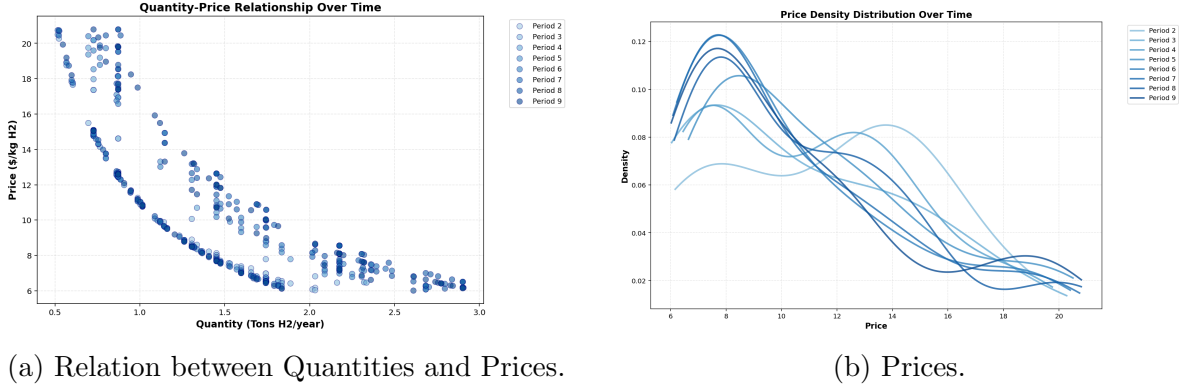


Figure 5: Distribution of Contracts Offered by Year.

Price dynamics exhibits predictable patterns throughout the adoption process. Our analysis reveals that median hydrogen prices decrease by about 33% from 2025 to 2035, with prices stabilizing as the market matures as shown in panel (b) of Figure 5 where prices go from \$16.22 kg/H<sub>2</sub> on average in 2026 to \$11.12 kg/H<sub>2</sub> in 2035. Price variability is highest in later periods (2032-2035), reflecting more heterogeneous contracts as the market expands. Figure 5 panel (a) shows the yearly quantities traded and the price for one kilogram of hydrogen established in the contracts at different years. During the same year, customers demanding more product should have a reduction in the unit price they pay by kilogram of hydrogen. As the technology improves, prices decrease overall but the differential in prices increases, meaning that consumers with higher demand get higher discounts as time passes, an advantage they get when signing long-term contracts earlier.

The early segments of the market show distinct characteristics that inform the targeting strategies. Figure 6 shows the profit share coming from customers with different

demand over time. The Figure indicates that high-demand segments (customers requiring 40+ tons of hydrogen annually) initially contribute disproportionately to supplier profits, around 45% of profits. However, this concentration decreases over time as network effects enable profitable service to lower-demand segments. This suggests a sequential targeting strategy starting with high-mileage fleet operators before expanding to broader market segments. Profits are increasing in time, the hydrogen manufacturer makes the most profits in period 9 where all the markets are active and the market share in each market is the largest.

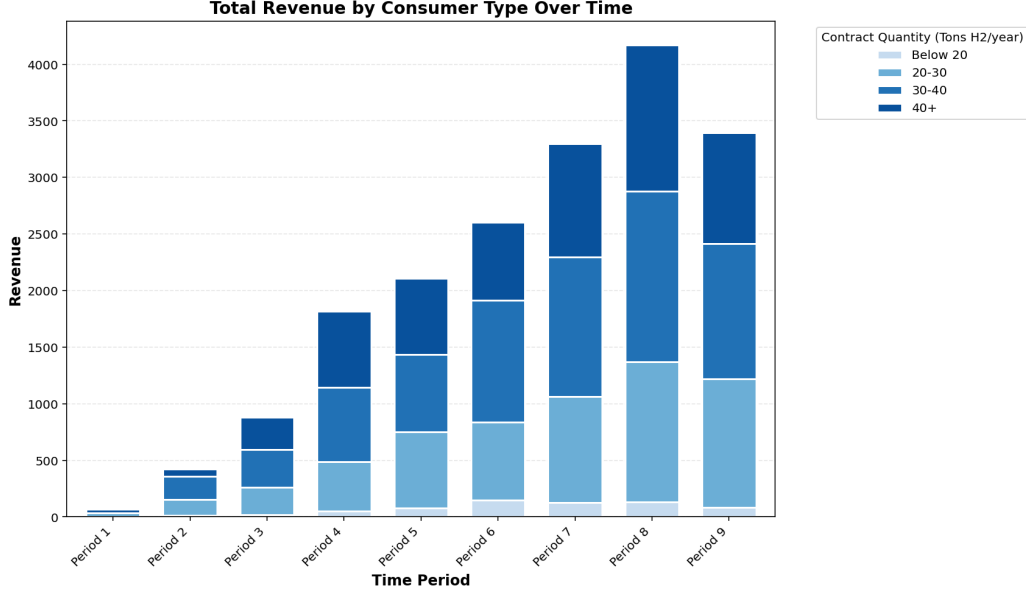


Figure 6: Profits by Demand Groups.

Network effects significantly influence the optimal entry strategy. Our simulations demonstrate that early entry losses in strategic markets can be justified by accelerated network effects. For instance, as shown in Table 4 entering market 11 in 2028 generates losses of \$4.5 million initially but cumulative profits exceeding \$34 million, so entry into the markets proves optimal despite short-term losses. This suggests hydrogen suppliers should prioritize establishing presence in high-potential markets even at initial loss positions.

Policy interventions differ in their effectiveness. Station subsidies primarily accelerate market penetration timing without significantly affecting final market share, while vehicle subsidies directly expand the market by approximately 9 percentage points. These findings indicate that policymakers should coordinate supply-side and demand-side incentives, with supply incentives deployed early in the adoption cycle and demand incentives structured to overcome initial adoption resistance.

## 7 Conclusion and Further Research.

This paper provides a framework for analyzing technology adoption under uncertainty, network effects, and policy interventions, with a focus on the adoption of hydrogen in the heavy-duty transportation sector. By calibrating the model with data from the New York Advanced Clean Fleets (ACF) reporting program and cost calibrations with proprietary data from Air Liquide, we have demonstrated the model’s applicability in estimating fuel demand and predicting market outcomes under various contractual and policy scenarios.

Our numerical results reveal several key insights for the hydrogen heavy-duty transportation sector. First, personalized contracting strategies dramatically outperform uniform pricing approaches, generating approximately 20 times higher profits (\$1.5 billion versus \$109 million over 10 years) while achieving substantially greater market penetration (42% versus 9.4% market share). Second, price dynamics follow predictable patterns, with median hydrogen prices declining by approximately 33% from 2026 to 2035 (from \$16.22/kg to \$12.13/kg) as technology costs fall and network effects strengthen. Third, optimal entry strategies require accepting early losses in strategic markets to capture long-term network benefits—for instance, entering certain markets generates initial losses of approximately \$4-7 million but yields cumulative profits exceeding \$34 million. Finally, policy interventions have differential impacts: station subsidies primarily accelerate adoption timing and increase profitability, while vehicle subsidies directly expand market share, in some scenarios reaching up to 60% penetration by 2035.

Future research could extend this work by incorporating additional dimensions of uncertainty into the model. For instance, the evolution of customer preferences over time or variability in government incentives could significantly influence adoption decisions. Integrating these uncertainties would not only make the model more realistic but also provide a richer understanding of potential market dynamics. Additionally, the model could explore interactions between regional markets, especially as network effects often extend beyond localized boundaries, influencing adoption rates in neighboring regions. Finally, we are considering ways to incorporate competition that can influence the outside option for different consumers.

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## Appendix A - Different Model Assumptions.

### Single Market Price.

We first look at the case where the company can only charge one price on a market as a benchmark for our study. Given a price  $p_{mt}$  at market  $m$  and time  $t$ , consumers of type  $\theta$  on that market would demand  $q_{mt}(\theta)$  such that

$$\frac{\partial}{\partial q} V_{mt}(q_{mt}(\theta), N_t, \theta) = p_{mt}$$

If

$$\begin{aligned} u_{mt}(\theta) &\geq \bar{u}_{mt}(\theta) \\ u_{mt}(\theta) &\geq u_{mt+1}(\theta) \end{aligned}$$

If  $V_{mt}(\cdot)$  is increasing in  $q$  and  $\theta$ , at any  $t$ , there exists  $\underline{\theta}_{mt}$  such that clients above that type buy the product. The threshold solves the following equation

$$V_{mt}(q_{mt}(\underline{\theta}_{mt}), N_t, \underline{\theta}_{mt}) - P_t - q_{mt}(\underline{\theta}_{mt})p_{mt} = 0$$

The firm then maximizes

$$\begin{aligned} \max_{\{1_{mt}, p_{mt}\}_{t=1, \dots, T}} & \left\{ \underbrace{\sum_{t=1}^T \sum_{m=1}^M A_{mt}(p_{mt} - c_t) \int_{\underline{\theta}_{mt}}^{\infty} q_{mt}(\theta) f_m(\theta) d\theta}_{\text{Flow revenue}} - \underbrace{\sum_{t=1}^T \sum_{m=1}^M 1_{mt} C_{mt}}_{\text{Initial fixed cost}} \right\} \\ \text{s.t. } & \frac{\partial}{\partial q} V_{mt}(q_{mt}(\theta), N_t, \theta) = p_{mt}, \quad \forall \theta, m, t \\ & N_t = \sum_{m=1}^M A_{mt} \end{aligned}$$

Let  $Q_{mt}(p_{mt}) = \int_{\underline{\theta}}^{\infty} q_{mt}^*(\theta) f_m(\theta) d\theta$ , notice this is a function of  $p_{mt}$  as the quantity demanded depends on the price set by the firm. Define also  $Q'_{mt}(p_{mt}) = \frac{\partial Q_{mt}(p_{mt})}{\partial p_{mt}}$ . Then, the profit maximizing price satisfies

$$p_{mt}^* = c_t - \frac{Q_{mt}(p_{mt}^*)}{Q'_{mt}(p_{mt}^*)}$$

In our numerical approach, we will solve for  $p_{mt}^*$  and compute the profits in order to solve the entry problem.

### Asymmetric Information.

In this section, we describe a model where the producer firm cannot observe the type of the mobility clients, so the optimal strategy for the firm relies on designing a menu of contracts such that the clients self-select and reveal their types. The problem of the firm now has an additional constraint:

$$V_{mt}(q_{mt}(\theta), N_t, \theta) - q_{mt}(\theta)p_{mt}(\theta) - P_t \geq V_{mt}(q_{mt}(\theta'), N_t, \theta) - q_{mt}(\theta')p_{mt}(\theta') - P_t; \quad \forall \theta' \in \text{supp}(F_m)$$

**Proposition 3.** Under asymmetric information, the optimal contracts  $\{(q_{mr}^*, N_r^*)\}_{r \geq t}$  satisfy

$$\frac{\partial}{\partial q} V_{mt}(q_{mt}^*(\theta), N_r^*, \theta) - c_t = 0$$

And

$$V_{mt}(q_{mt}^*(\theta), N_t^*, \theta) - P_t 1_{mt}^* = p_{mt}^*(\theta) q_{mt}^*(\theta) + \int_{\underline{\theta}}^{\theta} \frac{\partial V_{mt}(q_{mt}^*(\theta), N_t^*, \theta)}{\partial \theta} d\theta$$

The buyer accepts the contract.

### Policy Uncertainty.

Policies that incentivize the adoption of new technologies, such as tax breaks for greener forms of transportation or funding for the production of new hydrogen plants, are uncertain.

$$\begin{aligned} & \max_{\{1_{mt}, (q_{mt}(\theta), p_{mt}(\theta))\}_{t=1, \dots, T}} \left\{ \underbrace{\sum_{t=1}^T \sum_{m=1}^M \int A_{mt} q_{mt}(\theta) (p_{mt}(\theta) - c_t) f_m(\theta) d\theta}_{\text{Flow revenue}} - \underbrace{\sum_{t=1}^T \sum_{m=1}^M 1_{mt} E[C_{mt} | Policy]}_{\text{Initial fixed cost}} \right\} \\ \text{s.t. } & u_{mt}(\theta) \geq \bar{u}_{mt}(\theta), \quad \forall \theta, m, t \\ & u_{mt}(\theta) \geq E[u_{m\tau}(\theta) | policy], \quad \forall \theta, m, \tau \geq t \\ & N_t = \sum_{m=1}^M A_{mt} \end{aligned}$$

Notes on how to simulate policy uncertainty: Solve the problem at time  $t$  using the expectations, get the value functions from the code we already have, then take the optimal action based on the solution and draw a policy shock, next period repeat the process with the new expectation given the policy shock.

## Appendix B - Proofs.

**Proof of proposition 1.** First notice that it is optimal for the firm to set the buyer's participation constraint to 0. If the contract is offered at time  $t$ , this implies

$$V_{mt}(q_{mt}^*(\theta), N_t^*) - P_t - \sum_{r=t}^T q_{mr}^* p_{mr}^*(\theta) + \sum_{r=t+1}^T V_{mr}(q_{mr}^*(\theta), N_r^*) = 0$$

$$\Rightarrow \sum_{r=t}^T q_{mr}^* p_{mr}^*(\theta) = \sum_{r=t}^T V_{mr}(q_{mr}^*(\theta), N_r^*) - P_t$$

If the firm waits to period  $t + 1$  to offer the contract

$$\sum_{r=t+1}^T q_{mr}^* p_{mr}^*(\theta) = \sum_{r=t+1}^T V_{mr}(q_{mr}^*(\theta), N_r^*) - P_{t+1}$$

The revenue from the contract starting at time  $t$  minus that of the contract starting at time  $t + 1$  is

$$q_{mt}^*(\theta) p_{mt}^*(\theta) = V_{mt}(q_{mt}^*(\theta), N_t^*) - P_t + P_{t+1}$$

So, offering the contract at time  $t$  is more profitable if  $V_{mt}(q_{mt}^*(\theta), N_t^*) - P_t + P_{t+1} > 0$ .

The second part of the proposition comes from the FOC of the firm after plugging in the revenue from the participation constraint. The buyers accept the contract by construction given it gives them at least their reservation utility.

Appendix C - Additional Tables and Figures.

Table 5: Linear model of Truck revenue on fuel demand and network size

|                         | <i>Dependent variable:</i>  |
|-------------------------|-----------------------------|
|                         | log(truck revenue)          |
| active regions          | 0.205***<br>(0.012)         |
| truck fuel demand       | 0.147***<br>(0.007)         |
| Weight Class 4-6        | 0.211***<br>(0.050)         |
| Weight Class 7-8        | 0.001<br>(0.048)            |
| Constant                | 0.835***<br>(0.046)         |
| Observations            | 4,454                       |
| R <sup>2</sup>          | 0.159                       |
| Adjusted R <sup>2</sup> | 0.158                       |
| Residual Std. Error     | 1.282 (df = 4449)           |
| F Statistic             | 210.561*** (df = 4; 4449)   |
| <i>Note:</i>            | *p<0.1; **p<0.05; ***p<0.01 |

Table 6: Summary of Parameter Calibration.

| Parameter              | Description                                                             | Value                                          | Source.                                                                                                                                                                |
|------------------------|-------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Parameters     |                                                                         |                                                |                                                                                                                                                                        |
| $T$                    | Model Horizon                                                           | 10                                             | Model projected for 2025 - 2035.                                                                                                                                       |
| $M$                    | Number of Markets                                                       | 5                                              | First two numbers of ZIP in the State of New York                                                                                                                      |
| Demand Side Parameters |                                                                         |                                                |                                                                                                                                                                        |
| $F_m(\theta)$          | Distribution of yearly demand by tractor ( $\theta$ ) in each market    | -                                              | Based on observed mileage in <a href="#">State of New York (2024)</a> and fuel economy of class 7-8 tractors of 0.145kgH2/mile ( <a href="#">Basma et al., 2023</a> ). |
| $\phi_m$               | Market Size: Number of potential trucks in each market.                 | -                                              | Number of trucks reported in <a href="#">State of New York (2024)</a> and <a href="#">BizMiner (2024)</a>                                                              |
| $P_0$                  | TCO for FCEV class 7-8 Tractors minus TCO for diesel class 7-8 Tractors | \$300,000-\$470,000                            | <a href="#">Hunter et al. (2021)</a> and <a href="#">State of New York (2024)</a> .                                                                                    |
| $\delta_t$             | Expected reduction in FCEV-Diesel TCO                                   | 5%                                             | <a href="#">Hunter et al. (2021)</a> , <a href="#">Basma and Rodríguez (2022)</a>                                                                                      |
| $\sigma^2$             | TCO shock variance                                                      | 0.5                                            | <a href="#">Hunter et al. (2021)</a> , <a href="#">Basma and Rodríguez (2022)</a>                                                                                      |
| Supply Side Parameters |                                                                         |                                                |                                                                                                                                                                        |
| $c_t$                  | Variable cost of hydrogen production.                                   | \$7-\$12/kg in 2025 to \$4-\$7/kg in 2035      | <a href="#">Basma et al. (2023)</a> and proprietary data accounting for energy costs, maintenance, operation, and transportation costs of liquid hydrogen.             |
| $C_{m0}$               | Fixed costs: EPC with capacity of 2 tons/day in 2023                    |                                                | <a href="#">Basma et al. (2023)</a>                                                                                                                                    |
| N. Stations            | Number of stations needed to supply each region.                        | $(\phi_m \times E_m[\theta]) / (2 \times 365)$ | <a href="#">State of New York (2024)</a>                                                                                                                               |
| $C_{mt}$               | Fixed costs: EPC with capacity of 2 tons/day                            | Around \$3-4 MM per station in 2030.           | Proprietary data and reporting data from <a href="#">State of New York (2024)</a> .                                                                                    |

Table 7: Example of One Organization Info on the Database.

| Org. ID | Fac. ID | Rev 2021     | Sust. Plan | Fac. State | Fac. ZIP | Fuel Type | Weight Class | Avg. Annual Miles | Frac. in NYS | # of Veh. | <100 mi | 100-150 mi | 150-200 mi | 200-300 mi | >300 mi |
|---------|---------|--------------|------------|------------|----------|-----------|--------------|-------------------|--------------|-----------|---------|------------|------------|------------|---------|
| 2       | 131073  | \$100-\$499M | No         | NY         | 12020    | Diesel    | 7-8          | 12620             | 1            | 19        | 8       | 5          | 3          | 3          | 0       |
| 2       | 131073  | \$100-\$499M | No         | NY         | 12020    | Gasoline  | 2b-3         | 11330             | 1            | 10        | 10      | 0          | 0          | 0          | 0       |
| 2       | 131074  | \$100-\$499M | No         | NY         | 13753    | Gasoline  | 4-6          | 3402              | 1            | 1         | 1       | 0          | 0          | 0          | 0       |
| 2       | 131075  | \$100-\$499M | No         | NY         | 12054    | Gasoline  | 2b-3         | 11846             | 1            | 1         | 1       | 0          | 0          | 0          | 0       |
| 2       | 131081  | \$100-\$499M | No         | NY         | 13411    | Diesel    | 7-8          | 10116             | 1            | 3         | 0       | 1          | 1          | 1          | 0       |
| 2       | 131081  | \$100-\$499M | No         | NY         | 13411    | Gasoline  | 4-6          | 731               | 1            | 1         | 1       | 0          | 0          | 0          | 0       |
| 2       | 131084  | \$100-\$499M | No         | NY         | 12901    | Diesel    | 7-8          | 6455              | 1            | 2         | 0       | 1          | 1          | 0          | 0       |
| 2       | 131084  | \$100-\$499M | No         | NY         | 12901    | Gasoline  | 2b-3         | 3093              | 1            | 1         | 0       | 1          | 0          | 0          | 0       |
| 2       | 131085  | \$100-\$499M | No         | NY         | 12804    | Gasoline  | 2b-3         | 2943              | 1            | 1         | 1       | 0          | 0          | 0          | 0       |
| 2       | 131085  | \$100-\$499M | No         | NY         | 12804    | Diesel    | 7-8          | 19511             | 1            | 1         | 1       | 0          | 0          | 0          | 0       |
| 2       | 131086  | \$100-\$499M | No         | NY         | 12977    | Diesel    | 7-8          | 16091             | 1            | 1         | 0       | 0          | 1          | 0          | 0       |
| 2       | 131086  | \$100-\$499M | No         | NY         | 12977    | Gasoline  | 4-6          | 5746              | 1            | 2         | 2       | 0          | 0          | 0          | 0       |